ALKALI ADSORPTION ON GRAPHENE

Validated PBE + DFT-D3(BJ) relaxation results • Current checkpoint: 15 Oct 2026

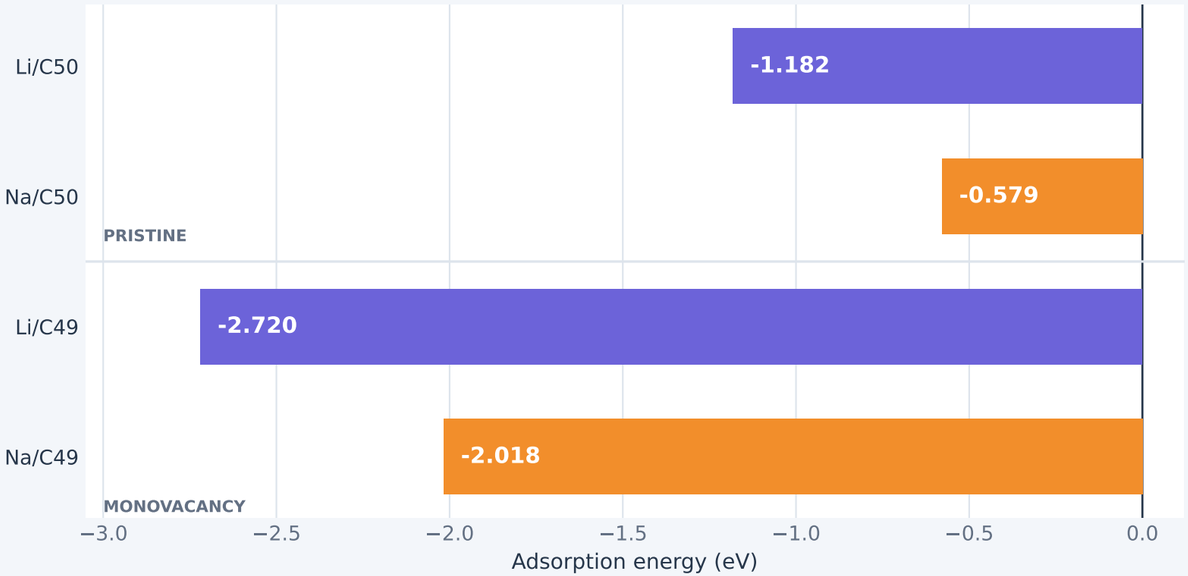
4 / 4

adsorption relaxations strictly converged

Adsorption energy

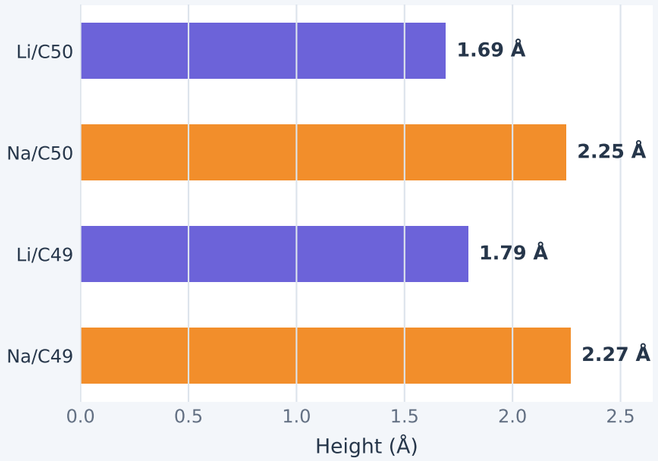
Atom-referenced Eads; more negative means stronger exothermic binding

RELAXED GEOMETRIES • 6×6×1



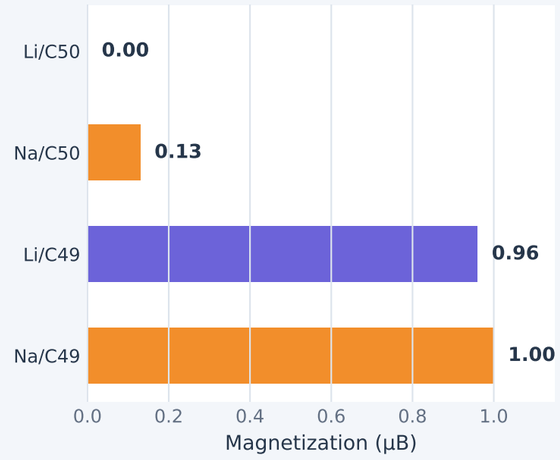
Adsorption height

Metal height above the mean carbon plane



Magnetism

Total magnetic moment per simulation cell



What the calculations show

01 **Vacancies strengthen binding**

Li: 1.538 eV stronger • Na: 1.439 eV stronger than pristine graphene

02 **Li binds more strongly than Na**

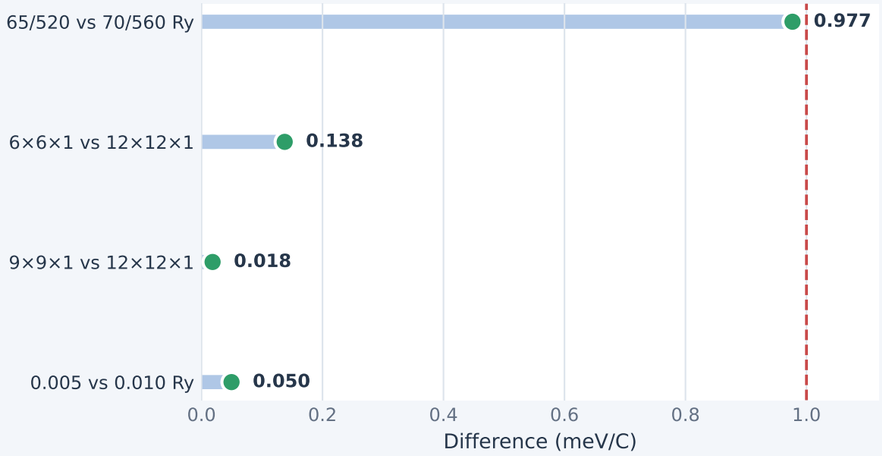
By 0.603 eV on pristine C50 and 0.702 eV at the C49 vacancy

03 **Vacancy magnetism persists**

Bare C49: 1.60 μB → Li/C49: 0.96 μB • Na/C49: 1.00 μB

Numerical validation

Energy difference per carbon atom; dashed line = 1 meV/C criterion



Quality gates

Relaxations **4 / 4 PASS**

Largest force component

0.00039186 Ry/Bohr

criterion ≤ 0.0004

9×9×1 final SCFs **RUNNING**

Na/C49 attempt02 • GPU active

METHOD PBE + DFT-D3(BJ) | PAW pseudopotentials | 65/520 Ry | 6×6×1 relaxation | MV smearing 0.01 Ry | assume_isolated = '2D'

Quantum ESPRESSO 7.6 • GPU